Lctr 9: LTI Eigenfunctions and the Continuous-Time Fourier Series

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Focus: Sections 3.0 – 3.3 (Pages 177 to 195)

9.1 The Big Picture: A New Way to Decompose Signals

Why This Matters

In Lecture 8, we learned that **any** discrete-time signal can be decomposed into shifted impulses, and that the output of a Linear Time-Invariant (LTI) system is found via **convolution**. That approach works, but now we introduce an alternative decomposition—using **complex exponentials** instead of impulses. This leads to the **Fourier series**, one of the most powerful tools in all of engineering. It lets us decompose a signal into its **frequency components**—just like a prism splits white light into a rainbow of colors.

Recall from Chapter 2: we decomposed signals as sums of shifted impulses. This led to convolution.

Now we ask: *what if we decompose signals as sums of complex exponentials instead?* This gives us a completely different—and equally powerful—way to analyze LTI systems.

Why complex exponentials? Because they are **eigenfunctions** of LTI systems. Don't worry—we will explain exactly what that means next.

9.2 Complex Exponentials Are Eigenfunctions of LTI Systems

9.2.1 What Is an Eigenfunction?

An **eigenfunction** of a system is a special input signal that, when passed through the system, comes out as itself multiplied by a constant. That constant is called the **eigenvalue**.

In plain language:

- You put a signal in $\rightarrow$ the same signal comes out, just scaled (possibly in both amplitude and phase).
- The shape of the signal does *not* change.

9.2.2 The Eigenfunction Property (Continuous Time)

For a continuous-time (CT) LTI system with impulse response $h(t)$, if the input is the complex exponential $x(t) = e^{st}$ (where s is a complex number), then the output is:

$$y(t) = H(s) \cdot e^{st}$$

The output is the *same* complex exponential e^{st}, just multiplied by a complex constant $H(s)$.

Where does $H(s)$ come from? It is computed from the impulse response $h(t)$:

$$H(s) = \int_{-\infty}^{+\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

Key Insight

The eigenfunction property means that if you input a complex exponential e^{st} into *any* LTI system, the output is $H(s)\,e^{st}$. The system only changes the amplitude and phase of the exponential—it never changes its frequency or shape. This is what makes complex exponentials so special for LTI analysis.

Numerical Example: Eigenfunction in Action

System: $h(t) = e^{-t}\,u(t)$ (a decaying exponential, causal)
Input: $x(t) = \cos(2t)$ (a cosine at $\omega = 2$ rad/s)
Step 1: Compute the system function at $s = j\omega = j2$:

$$H(j2) = \int_0^\infty e^{-\tau}\,e^{-j2\tau}\,d\tau = \int_0^\infty e^{-(1+j2)\tau}\,d\tau = \frac{1}{1+j2}$$

Step 2: Convert to polar form (magnitude and phase):

$$|H(j2)| = \frac{1}{\sqrt{1^2+2^2}} = \frac{1}{\sqrt{5}} \approx 0.447, \qquad \angle H(j2) = -\arctan\left(\frac{2}{1}\right) \approx -63.4$$

Step 3: Write the output. Since $\cos(2t)$ is the real part of e^{j2t}:

$$y(t) = 0.447\,\cos(2t - 63.4)$$

What happened? The cosine went in at amplitude 1. It came out at amplitude 0.447 and shifted by -63.4. The frequency (2 rad/s) did *not* change. That is the eigenfunction property!

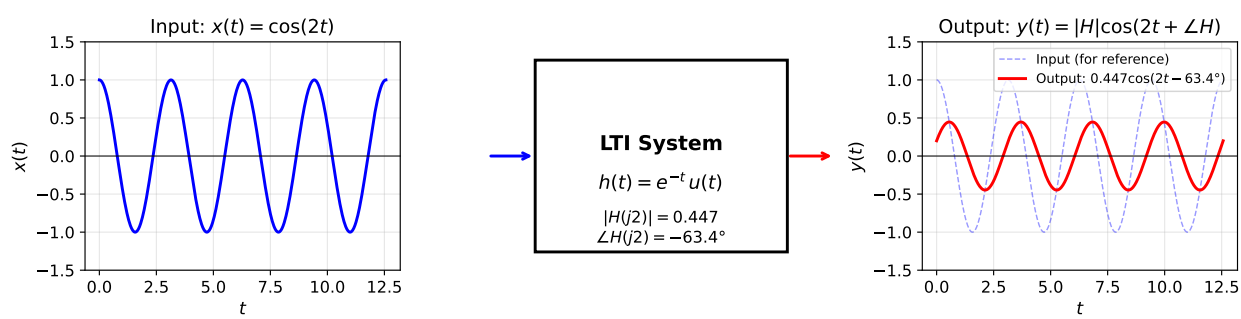

Figure 1: The eigenfunction property in action: a cosine at $\omega = 2$ rad/s enters the LTI system $h(t) = e^{-t}u(t)$ and exits as the *same* cosine, only scaled to 0.447 and shifted by -63.4. The frequency does not change.

9.2.3 Another Example: Time-Delay System

Numerical Example: Delay System as Eigenfunction

System: $y(t) = x(t-3)$ (delays input by 3 seconds), so $h(t) = \delta(t-3)$.
Input: $x(t) = e^{j2t}$
Step 1: Compute $H(s) = e^{-3s}$. At $s = j2$: $H(j2) = e^{-j6}$.
Step 2: This has $|H(j2)| = 1$ (no amplitude change) and $\angle H(j2) = -6$ rad.
Step 3: Output: $y(t) = e^{-j6} \cdot e^{j2t} = e^{j2(t-3)}$.
Check: A delay by 3 simply shifts e^{j2t} to $e^{j2(t-3)}$—exactly what we expect!

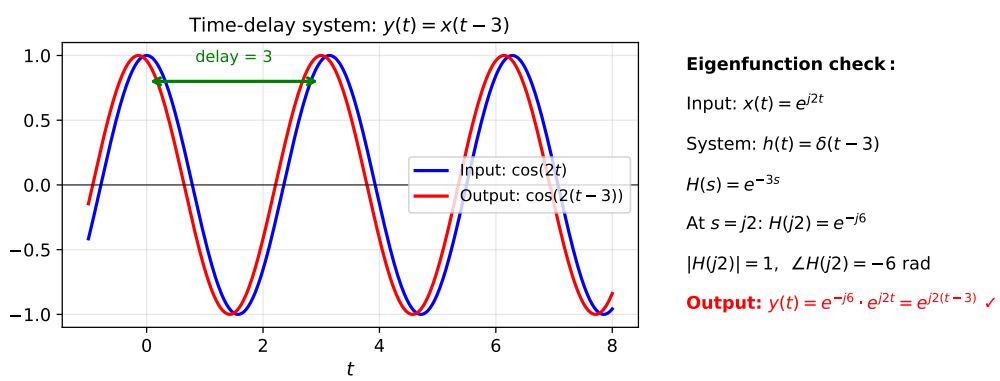

Figure 2: A time-delay system is an LTI system. The eigenvalue $H(j2) = e^{-j6}$ has magnitude 1 (no scaling) and phase -6 rad (pure time shift). The cosine comes out delayed but otherwise unchanged.

9.2.4 The Eigenfunction Property (Discrete Time)

The exact same idea works in discrete time (DT). For a DT LTI system with impulse response $h[n]$, if the input is $x[n] = z^n$ (where z is a complex number), then the output is:

$$y[n] = H(z) \cdot z^n, \qquad \text{where } H(z) = \sum_{k=-\infty}^{+\infty} h[k]\, z^{-k}$$

Quick DT Eigenfunction Example

System: $y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1]$ (two-point average), so $h[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1]$.
Input: $x[n] = e^{j\pi n/3}$ (DT exponential at $\omega = \pi/3$)
Step 1: $H(z) = \frac{1}{2} + \frac{1}{2}z^{-1}$. At $z = e^{j\pi/3}$:

$$H(e^{j\pi/3}) = \frac{1}{2} + \frac{1}{2}e^{-j\pi/3} = \frac{1}{2} + \frac{1}{2}\left(\frac{1}{2} - j\frac{\sqrt{3}}{4}\right) = \frac{3}{4} - j\frac{\sqrt{3}}{4}$$

$|H| = \sqrt{(3/4)^2 + (\sqrt{3}/4)^2} = \sqrt{9/16 + 3/16} = \sqrt{3/4} \approx 0.866$
Output: $y[n] = 0.866\, e^{j(\pi n/3 + \angle H)}$ — same frequency, just scaled and shifted!

9.2.5 Why Does This Help Us?

Here is the crucial consequence. Suppose we can write an input signal as a **sum of complex exponentials**:

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + a_3 e^{s_3 t}$$

Then, by the **superposition property** (because the system is linear), the output is:

$$y(t) = a_1 H(s_1) e^{s_1 t} + a_2 H(s_2) e^{s_2 t} + a_3 H(s_3) e^{s_3 t}$$

Each component passes through independently, and each one gets multiplied by the system's response at that particular frequency.

Key Insight

The strategy: (1) Decompose the input into complex exponentials. (2) Multiply each component by the system's eigenvalue at that frequency. (3) Add them back up to get the output. This is the foundation of **Fourier analysis**.

9.3 Periodic Signals and Harmonically Related Complex Exponentials

9.3.1 Review: What Makes a Signal Periodic?

A continuous-time signal $x(t)$ is **periodic** with period T if:

$$x(t) = x(t+T) \quad \text{for all } t$$

The **fundamental period** T is the smallest positive value for which this holds, and the **fundamental frequency** is:

$$\omega_0 = \frac{2\pi}{T} \quad \text{(in radians per second, abbreviated rad/s)}$$

Numerical Example: Period and Frequency

If a signal repeats every $T = 0.01$ seconds (like the power grid frequency):

$$\omega_0 = \frac{2\pi}{0.01} = 200\pi \approx 628.3 \text{ rad/s}, \qquad f_0 = \frac{1}{T} = 100 \text{ Hz (Hertz)}$$

If a signal repeats every $T = 1$ second:

$$\omega_0 = \frac{2\pi}{1} = 2\pi \approx 6.28 \text{ rad/s}, \qquad f_0 = 1 \text{ Hz}$$

Remember: ω_0 (rad/s) $= 2\pi \cdot f_0$ (Hz). Always check your units!

9.3.2 Harmonically Related Complex Exponentials

The set of complex exponentials that share the same period T is:

$$\phi_k(t) = e^{jk\omega_0 t} = e^{jk(2\pi/T)t}, \quad k = 0, \pm 1, \pm 2, \ldots$$

Key terminology:

- $k = 0$: This gives $\phi_0(t) = 1$ (a constant). This is the **DC component** ("Direct Current"—a term borrowed from electrical engineering meaning a constant, non-oscillating value).
- $k = \pm 1$: These have frequency ω_0. Together they form the **fundamental component** (also called the **first harmonic**).
- $k = \pm 2$: Frequency $2\omega_0$. This is the **second harmonic**.
- $k = \pm N$: Frequency $N\omega_0$. This is the N**th harmonic**.

The word "harmonic" comes from music: harmonics are the overtones that give instruments their distinct sound.

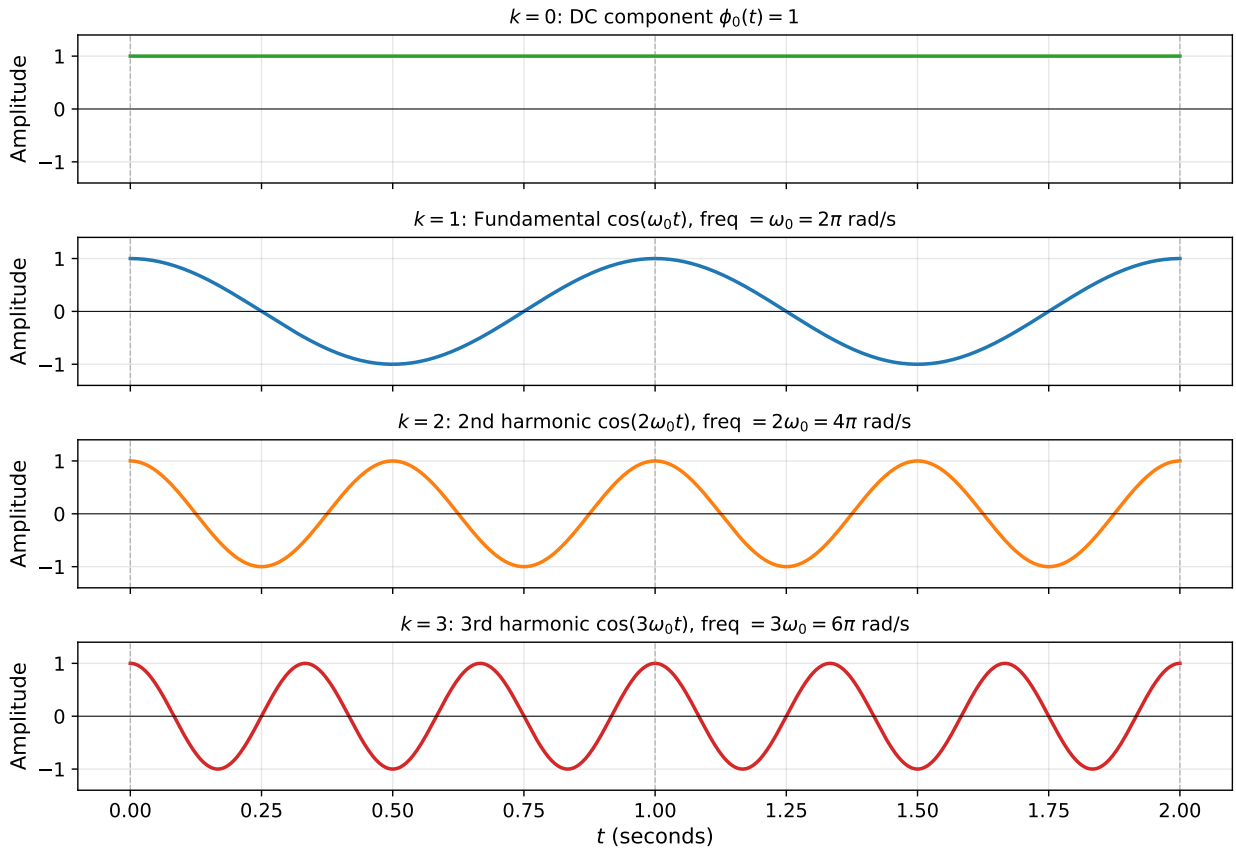

Figure 3: The first four harmonics for a signal with $T = 1$ s ($\omega_0 = 2\pi$ rad/s). The DC component ($k = 0$) is a constant. Each successive harmonic oscillates faster: the kth harmonic completes exactly k full cycles per period. Dashed lines mark period boundaries.

Numerical Example: Harmonic Frequencies with T

For $T = 1$ s, the fundamental frequency is $\omega_0 = 2\pi$ rad/s. The harmonics are:

k	Frequency (rad/s)	Frequency (Hz)	Name
0	0	0	DC (constant)
± 1	$2\pi \approx 6.28$	1	Fundamental (1st harmonic)
± 2	$4\pi \approx 12.57$	2	2nd harmonic
± 3	$6\pi \approx 18.85$	3	3rd harmonic
± 5	$10\pi \approx 31.42$	5	5th harmonic

Each harmonic oscillates exactly $|k|$ times per period (see Figure 3).

9.3.3 Building Periodic Signals from Harmonics

A linear combination of these harmonically related exponentials is:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t}$$

This is called the **Fourier series representation** of $x(t)$. Every term in this sum is periodic with period T, so their sum is also periodic with period T.

The coefficients a_k are complex numbers that tell us *how much* of each harmonic is present in the signal.

Key Insight

Fourier's remarkable claim: Essentially any periodic signal of practical interest can be written as a sum of harmonically related complex exponentials. This means we can break down even complicated-looking periodic waveforms (square waves, sawtooth waves, etc.) into simple sinusoidal building blocks.

9.4 The Fourier Series: Synthesis and Analysis Equations

9.4.1 The Two Key Equations

The Fourier series of a periodic signal $x(t)$ with period T and fundamental frequency $\omega_0 = 2\pi/T$ is defined by two equations:

Synthesis equation (how to build the signal from its components):

$$\boxed{x(t) = \sum_{k=-\infty}^{+\infty} a_k \, e^{jk\omega_0 t}}$$

Analysis equation (how to find the coefficients from the signal):

$$\boxed{a_k = \frac{1}{T} \int_T x(t) \, e^{-jk\omega_0 t} \, dt}$$

where $\int_T$ means "integrate over any interval of length T" (for example, from 0 to T, or from $-T/2$ to $T/2$—whichever is convenient).

Notation: The coefficients $\{a_k\}$ are called the **Fourier series coefficients** or **spectral coefficients**.

9.4.2 Understanding the Analysis Equation via Orthogonality

Where does the analysis equation come from? The idea is to use the **orthogonality** of complex exponentials. Here is the key mathematical fact:

$$\int_T e^{j(k-n)\omega_0 t}\, dt = \begin{cases} T, & \text{if } k = n \\ 0, & \text{if } k \neq n \end{cases}$$

In words: when we integrate two different harmonics multiplied together over one period, the result is zero (they "cancel out"). Only when the harmonics are the same do we get a nonzero result. This is what allows us to isolate each coefficient a_k individually.

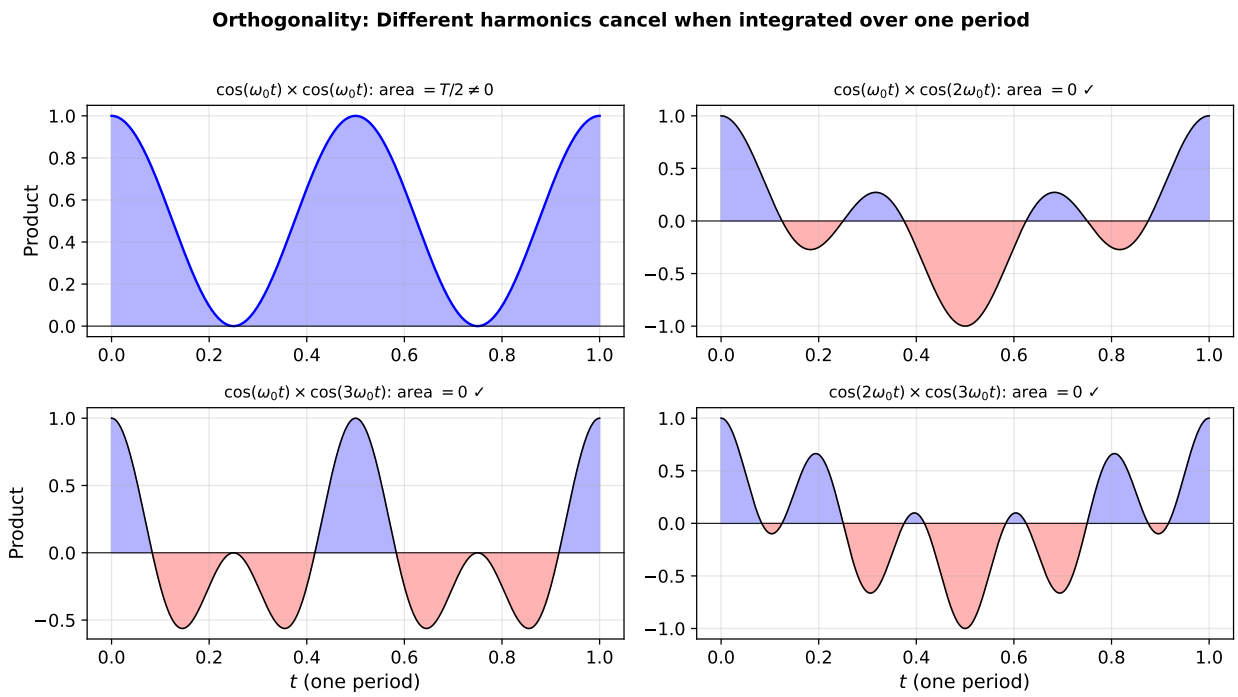

Figure 4: Visualizing orthogonality. When you multiply two *different* harmonics and integrate over one period, the positive (blue) and negative (red) areas cancel exactly, giving zero. When you multiply a harmonic by *itself*, the product is always positive, giving a nonzero integral ($T/2$ for real cosines). This is how the analysis equation "picks out" each coefficient individually.

Step-by-step derivation:

1. Start with $x(t) = \sum_k a_k e^{jk\omega_0 t}$.
2. Multiply both sides by $e^{-jn\omega_0 t}$:

$$x(t)\, e^{-jn\omega_0 t} = \sum_k a_k\, e^{j(k-n)\omega_0 t}$$

3. Integrate both sides over one period T:

$$\int_T x(t)\, e^{-jn\omega_0 t}\, dt = \sum_k a_k \underbrace{\int_T e^{j(k-n)\omega_0 t}\, dt}_{\text{equals } T \text{ if } k=n\text{, else } 0}$$

4. Only the $k = n$ term survives on the right side, giving $T \cdot a_n$.

5. Divide both sides by T:

$$a_n = \frac{1}{T} \int_T x(t)\, e^{-jn\omega_0 t}\, dt$$

9.4.3 Special Case: The DC Coefficient

When $k = 0$, the analysis equation simplifies to:

$$a_0 = \frac{1}{T} \int_T x(t)\, dt$$

This is simply the **average value** of the signal over one period. It is the "DC" (constant) part of the signal.

Numerical Example: DC Coefficient Is the Average

Consider a signal that over one period ($T = 4$ s) looks like: $x(t) = 3$ for $0 < t < 1$ and $x(t) = -1$ for $1 < t < 4$.
Step 1: Compute a_0:

$$a_0 = \frac{1}{4}\left[\int_0^1 3\, dt + \int_1^4 (-1)\, dt\right] = \frac{1}{4}\left[3(1) + (-1)(3)\right] = \frac{1}{4}(3-3) = 0$$

Interpretation: The average value of this signal over one period is zero—the positive and negative parts balance out, so there is no DC component.
Now change it to $x(t) = 3$ for $0 < t < 2$ and $x(t) = -1$ for $2 < t < 4$:

$$a_0 = \frac{1}{4}\left[3(2) + (-1)(2)\right] = \frac{1}{4}(6-2) = 1$$

Now the signal spends more time positive, so the DC component (average) is 1.

9.5 Worked Examples

9.5.1 Example 1: $x(t) = \sin(\omega_0 t)$

Goal: Find the Fourier series coefficients of $x(t) = \sin(\omega_0 t)$.
Step 1: Use Euler's relation to expand the sine:

$$\sin(\omega_0 t) = \frac{1}{2j}\, e^{j\omega_0 t} - \frac{1}{2j}\, e^{-j\omega_0 t}$$

Euler's relation states that $e^{j\theta} = \cos\theta + j\sin\theta$. From this, we can derive:

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

Step 2: Compare with the synthesis equation $x(t) = \sum_k a_k e^{jk\omega_0 t}$. By inspection:

$$a_1 = \frac{1}{2j} = -\frac{j}{2}, \quad a_{-1} = -\frac{1}{2j} = \frac{j}{2}, \quad a_k = 0 \text{ for } |k| \neq 1$$

Step 3 (sanity checks):

- $|a_1| = \frac{1}{2}$, $\angle a_1 = -90$; $\quad |a_{-1}| = \frac{1}{2}$, $\angle a_{-1} = +90$.
- Check conjugate symmetry: $a_{-1} = a_1^* = \left(-\frac{j}{2}\right)^* = \frac{j}{2}$. ✓
- Check: $a_1 e^{j\omega_0 t} + a_{-1} e^{-j\omega_0 t} = -\frac{j}{2} e^{j\omega_0 t} + \frac{j}{2} e^{-j\omega_0 t} = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j} = \sin(\omega_0 t)$. ✓

Result: A pure sine wave has only two nonzero Fourier coefficients, at $k = +1$ and $k = -1$. This makes intuitive sense—a sine wave is already a single-frequency signal, so it needs only one harmonic.

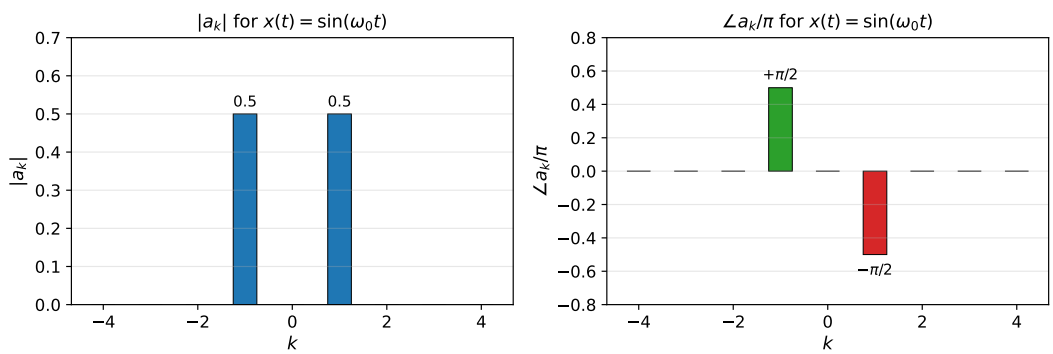

Figure 5: Fourier coefficients of $x(t) = \sin(\omega_0 t)$. **Left:** The magnitude spectrum $|a_k|$ shows two spikes at $k = \pm 1$, each of height $\frac{1}{2}$. **Right:** The phase spectrum shows $\angle a_{-1} = +\pi/2$ and $\angle a_1 = -\pi/2$ (antisymmetric, as expected for a sine).

9.5.2 Example 2: A Signal with Multiple Harmonics

Let $x(t) = 1 + \sin(\omega_0 t) + 2\cos(\omega_0 t) + \cos(2\omega_0 t + \pi/4)$, with fundamental frequency ω_0.

Step 1: Expand each term using Euler's relation:

$$\begin{aligned}
1 &= 1 \quad \text{(DC term)} \\
\sin(\omega_0 t) &= \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t} = -\frac{j}{2} e^{j\omega_0 t} + \frac{j}{2} e^{-j\omega_0 t} \\
2\cos(\omega_0 t) &= e^{j\omega_0 t} + e^{-j\omega_0 t} \\
\cos(2\omega_0 t + \tfrac{\pi}{4}) &= \frac{1}{2} e^{j\pi/4} e^{j2\omega_0 t} + \frac{1}{2} e^{-j\pi/4} e^{-j2\omega_0 t}
\end{aligned}$$

Step 2: Collect terms by harmonic. Group all terms multiplying $e^{jk\omega_0 t}$ for each k:

$$
\begin{aligned}
k = 0: &\quad a_0 = 1 \\
k = +1: &\quad a_1 = \underbrace{1}_{\text{from } 2\cos} + \underbrace{\left(-\frac{j}{2}\right)}_{\text{from sin}} = 1 - \frac{j}{2} \\
k = -1: &\quad a_{-1} = 1 + \frac{j}{2} \\
k = +2: &\quad a_2 = \frac{1}{2}\, e^{j\pi/4} = \frac{1}{2}\left(\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}\right) \approx 0.354 + 0.354j \\
k = -2: &\quad a_{-2} = \frac{1}{2}\, e^{-j\pi/4} \approx 0.354 - 0.354j \\
|k| > 2: &\quad a_k = 0
\end{aligned}
$$

Step 3 (compute magnitudes and phases):

$$
\begin{aligned}
&|a_1| = \sqrt{1^2 + (1/2)^2} = \sqrt{5/4} \approx 1.118, \quad \angle a_1 = \arctan\left(\frac{-1/2}{1}\right) \approx -26.6 \\
&|a_2| = 1/2 = 0.500, \quad \angle a_2 = \pi/4 = 45
\end{aligned}
$$

Step 4 (verify conjugate symmetry): Since $x(t)$ is real, we must have $a_{-k} = a_k^*$:

$$
a_{-1} = 1 + \frac{j}{2} = \left(1 - \frac{j}{2}\right)^* = a_1^* \quad \checkmark \qquad a_{-2} = \frac{1}{2}e^{-j\pi/4} = \left(\frac{1}{2}e^{j\pi/4}\right)^* = a_2^* \quad \checkmark
$$

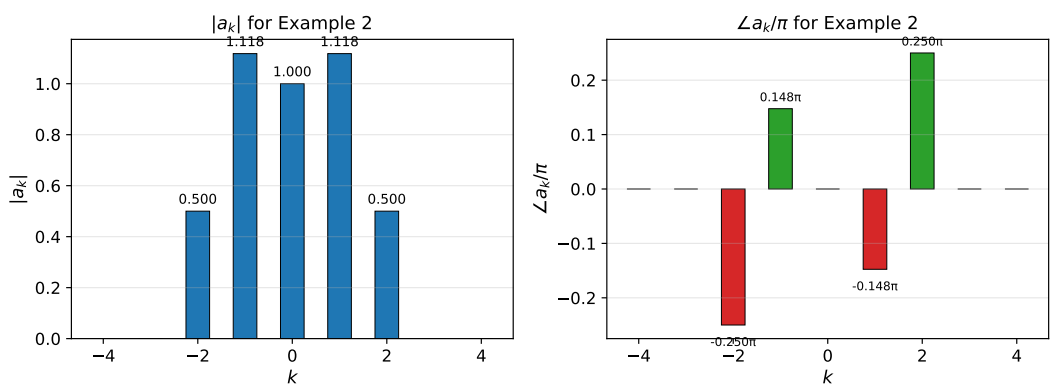

Figure 6: Fourier coefficients for Example 2. **Left:** Magnitude spectrum $|a_k|$—the $k = \pm 1$ components dominate. **Right:** Phase spectrum $\angle a_k/\pi$. Note the conjugate symmetry: $|a_k| = |a_{-k}|$ and $\angle a_k = -\angle a_{-k}$.

Key Insight

Trick for simple signals: When $x(t)$ is given as a sum of sines and cosines, you can often find the Fourier coefficients by **inspection**—just expand everything using Euler's relation and read off the coefficients. No integration needed!

9.5.3 Example 3: The Periodic Square Wave (with Numbers)

Consider the periodic square wave with period $T = 4$ and $T_1 = 1$, defined over one period as:

$$x(t) = \begin{cases} 1, & |t| < 1 \\ 0, & 1 < |t| < 2 \end{cases}$$

This signal is 1 for 2 seconds (from $t = -1$ to $t = 1$) centered at $t = 0$, and 0 for the remaining 2 seconds.

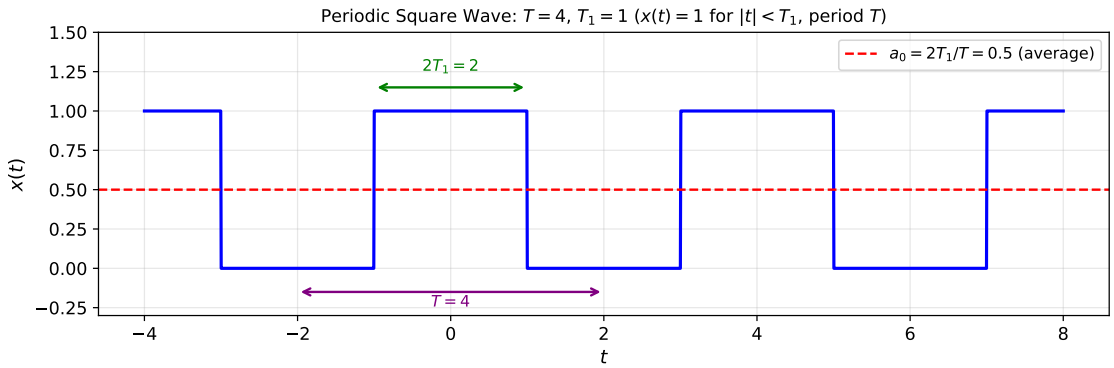

Figure 7: The periodic square wave with $T = 4$, $T_1 = 1$. The signal equals 1 for a duration of $2T_1 = 2$ seconds each period. The red dashed line shows the average value $a_0 = 0.5$.

Step 1: Find a_0 (the DC coefficient / average).
With $T = 4$, $\omega_0 = 2\pi/4 = \pi/2$ rad/s:

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\,dt = \frac{1}{4}\int_{-1}^{1} 1\,dt = \frac{1}{4}(2) = \frac{1}{2} = 0.5$$

Interpretation: The signal is "on" ($x = 1$) for half the period, so the average is $\frac{1}{2}$.
Step 2: Find a_k for $k \neq 0$.

$$\begin{aligned} a_k &= \frac{1}{T}\int_{-T_1}^{T_1} 1 \cdot e^{-jk\omega_0 t}\,dt = \frac{1}{4}\int_{-1}^{1} e^{-jk(\pi/2)t}\,dt \\ &= \frac{1}{4}\left[\frac{e^{-jk(\pi/2)t}}{-jk(\pi/2)}\right]_{-1}^{1} = \frac{1}{4}\cdot\frac{e^{-jk\pi/2} - e^{jk\pi/2}}{-jk\pi/2} \\ &= \frac{1}{4}\cdot\frac{-2j\sin(k\pi/2)}{-jk\pi/2} = \frac{1}{4}\cdot\frac{2\sin(k\pi/2)}{k\pi/2} = \frac{\sin(k\pi/2)}{k\pi} \end{aligned}$$

Step 3: Plug in actual numbers. Let's compute the first several coefficients:

k	$\sin(k\pi/2)$	$a_k = \frac{\sin(k\pi/2)}{k\pi}$	Decimal
0	—	$a_0 = 1/2$ (from average)	0.500
± 1	$\sin(\pi/2) = 1$	$1/\pi$	0.318
± 2	$\sin(\pi) = 0$	0	0.000
± 3	$\sin(3\pi/2) = -1$	$-1/(3\pi)$	-0.106
± 4	$\sin(2\pi) = 0$	0	0.000
± 5	$\sin(5\pi/2) = 1$	$1/(5\pi)$	0.064
± 6	$\sin(3\pi) = 0$	0	0.000
± 7	$\sin(7\pi/2) = -1$	$-1/(7\pi)$	-0.045

Observations:

- All even harmonics ($k = \pm 2, \pm 4, \pm 6, \ldots$) are **zero**! This is a property of this particular square wave (because of its symmetry).
- The odd harmonics decay like $1/(k\pi)$—they get smaller as k increases, but never reach zero.
- The coefficients alternate in sign: $+, -, +, -, \ldots$ (for the odd harmonics).

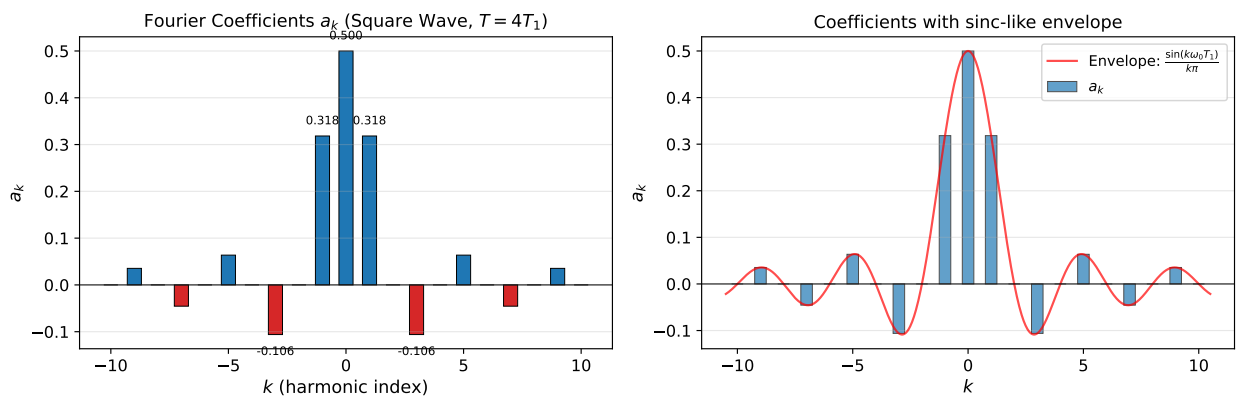

Figure 8: **Left:** The Fourier coefficients a_k of the square wave ($T = 4T_1$). Blue bars are positive, red bars are negative. Key values are annotated. **Right:** The coefficients sit on a sinc-like envelope $\frac{\sin(k\omega_0 T_1)}{k\pi}$ (red curve). The envelope crosses zero at even multiples of k, explaining why even harmonics vanish.

Common Mistake

When computing a_k for the square wave (or any signal with a $\sin(k \cdot \text{something})/(k \cdot \text{something})$ pattern), remember that a_0 must be computed **separately** using the DC formula $a_0 = \frac{1}{T}\int_T x(t)\,dt$. You **cannot** simply plug $k = 0$ into the $\sin/k$ formula because that creates a $0/0$ indeterminate form.

9.5.4 Seeing Fourier Series Build a Square Wave

Now let's *see* what happens when we add up the harmonics one by one. The partial sum with N harmonics is:

$$x_N(t) = a_0 + \sum_{k=1}^{N} \left[a_k\, e^{jk\omega_0 t} + a_{-k}\, e^{-jk\omega_0 t}\right] = \frac{1}{2} + \frac{2}{\pi}\cos(\omega_0 t) - \frac{2}{3\pi}\cos(3\omega_0 t) + \frac{2}{5\pi}\cos(5\omega_0 t) - \cdots$$

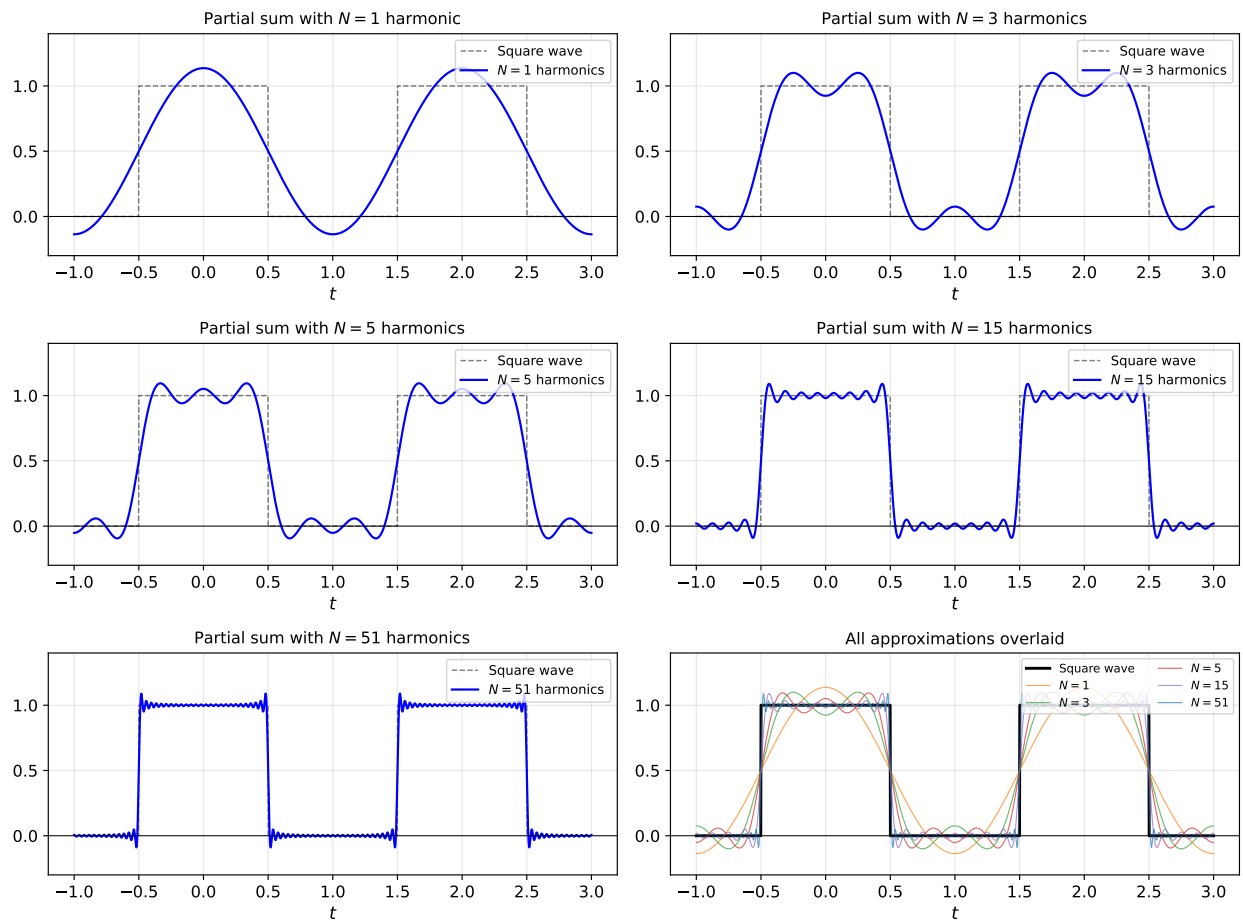

Figure 9: Building a square wave by adding harmonics. With $N = 1$, the approximation is just a shifted cosine. As N increases, the approximation improves and looks more and more like the actual square wave (dashed black). Even at $N = 51$ harmonics, there are small oscillations near the discontinuities—this is the **Gibbs phenomenon**, which we will study in Lecture 10.

Key Insight

What Figure 9 tells us: The Fourier series truly works! Even a sharp, discontinuous square wave can be built from smooth cosines. The more harmonics we add, the better the approximation. This is the power of Fourier analysis: any periodic signal can be decomposed into—and reconstructed from—its harmonic components.

9.6 Relationship Between Complex Exponential and Trigonometric Forms

For **real-valued** periodic signals, the Fourier coefficients satisfy the **conjugate symmetry** property:

$$a_{-k} = a_k^*$$

where $(\cdot)^*$ denotes the complex conjugate (flip the sign of the imaginary part).

This means the Fourier series can also be written in purely real, trigonometric forms. If we write a_k in polar form as $a_k = A_k\, e^{j\theta_k}$, then:

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k)$$

Alternatively, writing $a_k = B_k + jC_k$ (rectangular form):

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} [B_k \cos(k\omega_0 t) - C_k \sin(k\omega_0 t)]$$

Numerical Example: Converting Between Forms

From Example 1: $x(t) = \sin(\omega_0 t)$ has $a_1 = -j/2$.
Polar form: $A_1 = |a_1| = 1/2, \quad \theta_1 = \angle a_1 = -\pi/2$.
Trigonometric form: $x(t) = a_0 + 2A_1 \cos(\omega_0 t + \theta_1) = 0 + 2 \cdot \frac{1}{2}\cos(\omega_0 t - \pi/2)$.
Since $\cos(\alpha - \pi/2) = \sin(\alpha)$: $x(t) = \sin(\omega_0 t)$. ✓

From Example 2: $a_1 = 1 - j/2$ has $A_1 = \sqrt{5}/2 \approx 1.118$ and $\theta_1 \approx -26.6 \approx -0.464$ rad.
Trigonometric form contribution from $k = 1$:

$$2A_1 \cos(\omega_0 t + \theta_1) = 2(1.118)\cos(\omega_0 t - 0.464) = 2.236\cos(\omega_0 t - 26.6)$$

This single cosine at the fundamental frequency replaces both the $\sin(\omega_0 t)$ and $2\cos(\omega_0 t)$ terms from the original signal.

Key Insight

For real signals, the complex exponential form and the trigonometric form carry **exactly the same information**. The complex form is mathematically more compact. The trigonometric form is sometimes easier to interpret physically: each term is a cosine at a harmonic frequency with a specific amplitude and phase.

9.7 Summary and Key Formulas

Concept	**Formula/Definition**
Eigenfunction property (CT)	$e^{st} \xrightarrow{\text{LTI}} H(s)\, e^{st}$
Eigenfunction property (DT)	$z^n \xrightarrow{\text{LTI}} H(z)\, z^n$
Fundamental frequency	$\omega_0 = \dfrac{2\pi}{T}$
Fourier series (synthesis)	$x(t) = \displaystyle\sum_{k=-\infty}^{+\infty} a_k\, e^{jk\omega_0 t}$
Fourier series (analysis)	$a_k = \dfrac{1}{T}\displaystyle\int_T x(t)\, e^{-jk\omega_0 t}\, dt$
DC coefficient	$a_0 = \dfrac{1}{T}\displaystyle\int_T x(t)\, dt$ (average value)
Conjugate symmetry (real $x(t)$)	$a_{-k} = a_k^*$
Trigonometric form	$x(t) = a_0 + 2\displaystyle\sum_{k=1}^{\infty} \lvert a_k\rvert \cos(k\omega_0 t + \angle a_k)$
Euler's relation	$e^{j\theta} = \cos\theta + j\sin\theta$

9.8 Common Mistakes to Avoid

1. **Forgetting that a_0 needs separate treatment:** Many formulas for a_k have k in the denominator and are undefined at $k = 0$. Always compute a_0 from the average-value formula.

2. **Confusing ω_0 and f_0:** The fundamental frequency can be expressed in rad/s ($\omega_0 = 2\pi/T$) or in Hz ($f_0 = 1/T$). They are related by $\omega_0 = 2\pi f_0$. Be consistent.

3. **Wrong integration limits:** The analysis equation integrates over *any* interval of length T. Choose limits that make the integral easiest (often $-T/2$ to $T/2$ for symmetric signals, or 0 to T).

4. **Dropping the $1/T$ factor:** Don't forget the $\frac{1}{T}$ in front of the analysis integral.

5. **Thinking the sum is finite:** For continuous-time Fourier series, the sum goes from $k = -\infty$ to $k = +\infty$ (infinitely many harmonics). Only simple signals like pure sines have finitely many nonzero coefficients.

6. **Forgetting the identity $1/(2j) = -j/2$:** When expanding $\sin(\omega_0 t)$ via Euler, the coefficient of $e^{j\omega_0 t}$ is $1/(2j) = -j/2$, *not* $j/2$. Multiply top and bottom by $-j$ to verify: $\frac{1}{2j} \cdot \frac{-j}{-j} = \frac{-j}{2}$.

Key Insight

Quick reference — Euler's identities:

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}, \qquad \frac{1}{2j} = -\frac{j}{2}$$

You will use these constantly. Memorize them!

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